

UNIT -2 CLASS ACTIVITY**1) Hospital Patient Admission Analysis:**

A hospital has collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates department workload, admission variability, and patient distribution. Justify your selection with evidence.

CODE:

```
library(ggplot2)
library(dplyr)
library(readr)
df <- read_csv("hospital_admissions_1.csv")
glimpse(df)
totals <- df %>%
  group_by(department) %>%
  summarise(total_admissions = sum(admissions)) %>%
  arrange(desc(total_admissions))

p1 <- ggplot(totals, aes(x = reorder(department, -total_admissions),
                             y = total_admissions,
                             fill = department)) +
  geom_col() +
  labs(title = "Total Admissions by Department (Workload)",
       x = "Department", y = "Total Admissions") +
  theme_minimal() +
  theme(legend.position = "none")

print(p1)

p2 <- ggplot(df, aes(x = department, y = admissions, fill = department)) +
```

```
geom_boxplot() +  
labs(title = "Monthly Admission Variability by Department",  
      x = "Department", y = "Monthly Admissions") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
print(p2)
```

```
p3 <- ggplot(df, aes(x = admissions, fill = department)) +  
  geom_histogram(bins = 10, color = "white") +  
  facet_wrap(~ department, scales = "free_y") +  
  labs(title = "Distribution of Monthly Admissions per Department",  
        x = "Admissions", y = "Frequency") +  
  theme_minimal() +  
  theme(legend.position = "none")
```

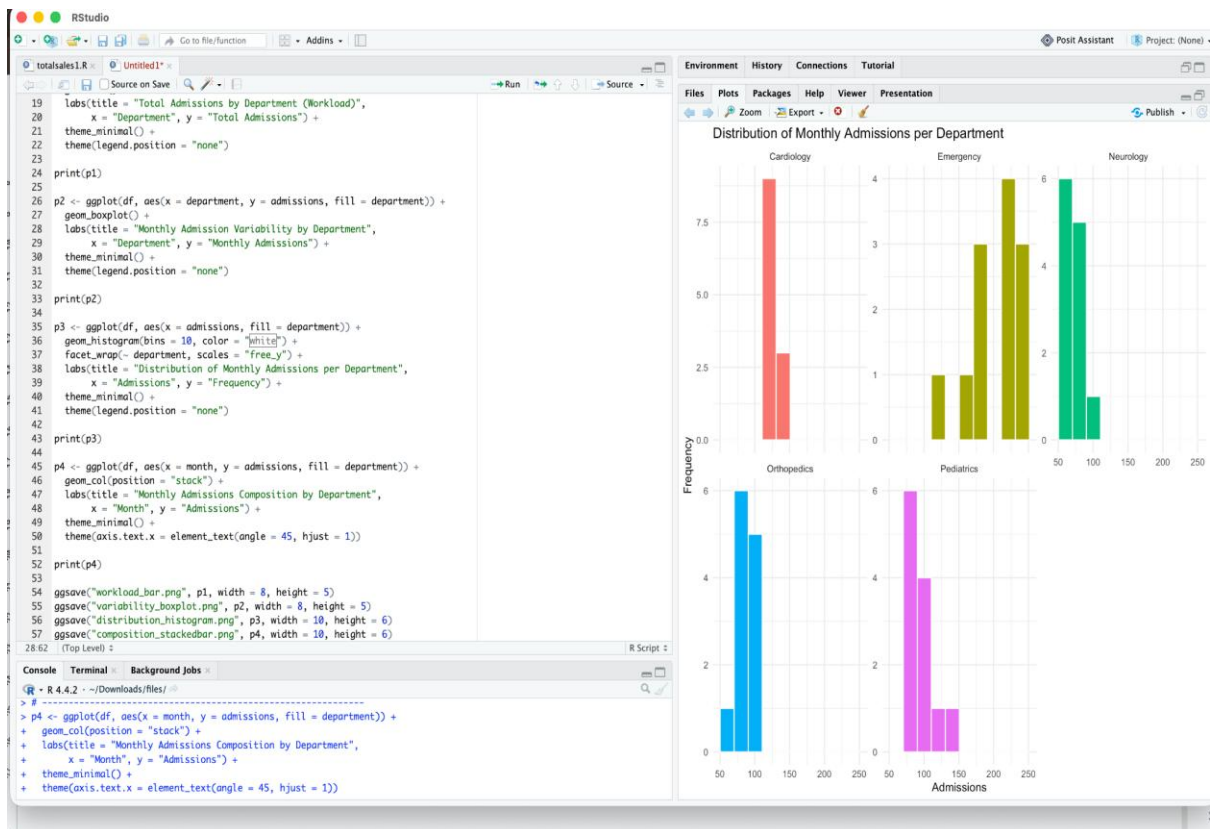
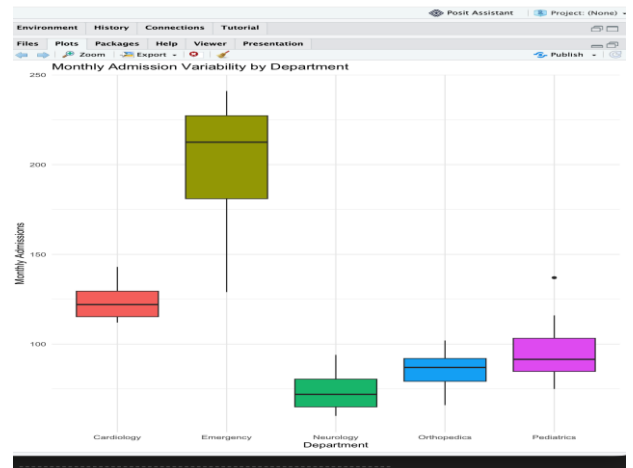
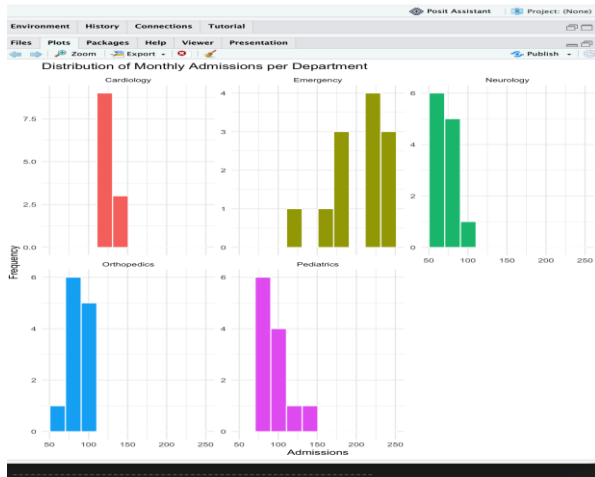
```
print(p3)
```

```
p4 <- ggplot(df, aes(x = month, y = admissions, fill = department)) +  
  geom_col(position = "stack") +  
  labs(title = "Monthly Admissions Composition by Department",  
        x = "Month", y = "Admissions") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
print(p4)
```

```
ggsave("workload_bar.png", p1, width = 8, height = 5)  
ggsave("variability_boxplot.png", p2, width = 8, height = 5)  
ggsave("distribution_histogram.png", p3, width = 10, height = 6)  
ggsave("composition_stackedbar.png", p4, width = 10, height = 6)
```

Screenshots:



2) E-Commerce Customer Spending Behaviour:

An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations (histogram, violin plot, box plot, density plot, bar chart). Evaluate which visualization best identifies high-value customers, spending variability, and purchasing trends. Recommend the most appropriate visualization for business executives and explain why.

CODE:

```
#
=====

=====
# E-Commerce Customer Spending Behaviour Analysis
# High-value customers | Spending variability | Purchasing trends
#
=====

=====

# Install packages once if you don't already have them:
# install.packages(c("ggplot2", "dplyr", "readr"))

library(ggplot2)
library(dplyr)
library(readr)

# -----
# 1. Load data
# (Change setwd() or use the full file path if R can't find the CSV)
# -----
# setwd("/Users/mohan/Documents") # uncomment and edit if needed
df <- read_csv("ecommerce_customers.csv")

glimpse(df)
summary(df$spending_amount)
```

```
# -----  
# 2. HISTOGRAM — overall shape of spending distribution  
# -----  
p1 <- ggplot(df, aes(x = spending_amount)) +  
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +  
  labs(title = "Distribution of Customer Spending Amounts",  
        x = "Spending Amount ($)", y = "Number of Customers") +  
  theme_minimal()
```

```
print(p1)
```

```
# -----  
# 3. VIOLIN PLOT — spending variability by category  
# -----  
p2 <- ggplot(df, aes(x = product_category, y = spending_amount, fill =  
product_category)) +  
  geom_violin(trim = FALSE) +  
  labs(title = "Spending Variability by Product Category (Violin Plot)",  
        x = "Product Category", y = "Spending Amount ($)") +  
  theme_minimal() +  
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust =  
1))
```

```
print(p2)
```

```
# -----  
# 4. BOX PLOT — identify high-value customers (outliers)  
# -----  
p3 <- ggplot(df, aes(x = product_category, y = spending_amount, fill =  
product_category)) +  
  geom_boxplot() +  
  labs(title = "Spending by Category — Box Plot (High-Value Customer  
Outliers)",  
        x = "Product Category", y = "Spending Amount ($)") +  
  theme_minimal() +  
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust =  
1))
```

```
print(p3)
```

```
# -----  
# 5. DENSITY PLOT — purchasing trends across categories  
# -----
```

```
p4 <- ggplot(df, aes(x = spending_amount, fill = product_category)) +  
  geom_density(alpha = 0.4) +  
  labs(title = "Spending Density by Product Category",  
        x = "Spending Amount ($)", y = "Density") +  
  theme_minimal()
```

```
print(p4)
```

```
# -----  
# 6. BAR CHART — total revenue by category (executive view)  
# -----
```

```
revenue_by_category <- df %>%  
  group_by(product_category) %>%  
  summarise(total_revenue = sum(spending_amount)) %>%  
  arrange(desc(total_revenue))
```

```
p5 <- ggplot(revenue_by_category, aes(x = reorder(product_category, -  
total_revenue),  
                                     y = total_revenue,  
                                     fill = product_category)) +  
  geom_col() +  
  labs(title = "Total Revenue by Product Category",  
        x = "Product Category", y = "Total Spending ($)") +  
  theme_minimal() +  
  theme(legend.position = "none")
```

```
print(p5)
```

```
# -----  
# 7. Identify high-value customers using IQR outlier rule  
# -----
```

```
Q1 <- quantile(df$spending_amount, 0.25)
```

```
Q3 <- quantile(df$spending_amount, 0.75)
```

```
IQR_val <- Q3 - Q1
```

```
upper_bound <- Q3 + 1.5 * IQR_val
```

```
high_value_customers <- df %>%
```

```
  filter(spending_amount > upper_bound) %>%
```

```
  arrange(desc(spending_amount))
```

```
cat("High-value threshold (Q3 + 1.5*IQR): $", round(upper_bound, 2), "\n")
```

```
cat("Number of high-value customers:", nrow(high_value_customers), "\n")
```

```
print(high_value_customers %>% select(customer_id, spending_amount,
product_category))
```

```
# -----
```

```
# 8. Save plots to files (optional)
```

```
# -----
```

```
ggsave("histogram_spending.png", p1, width = 8, height = 5)
```

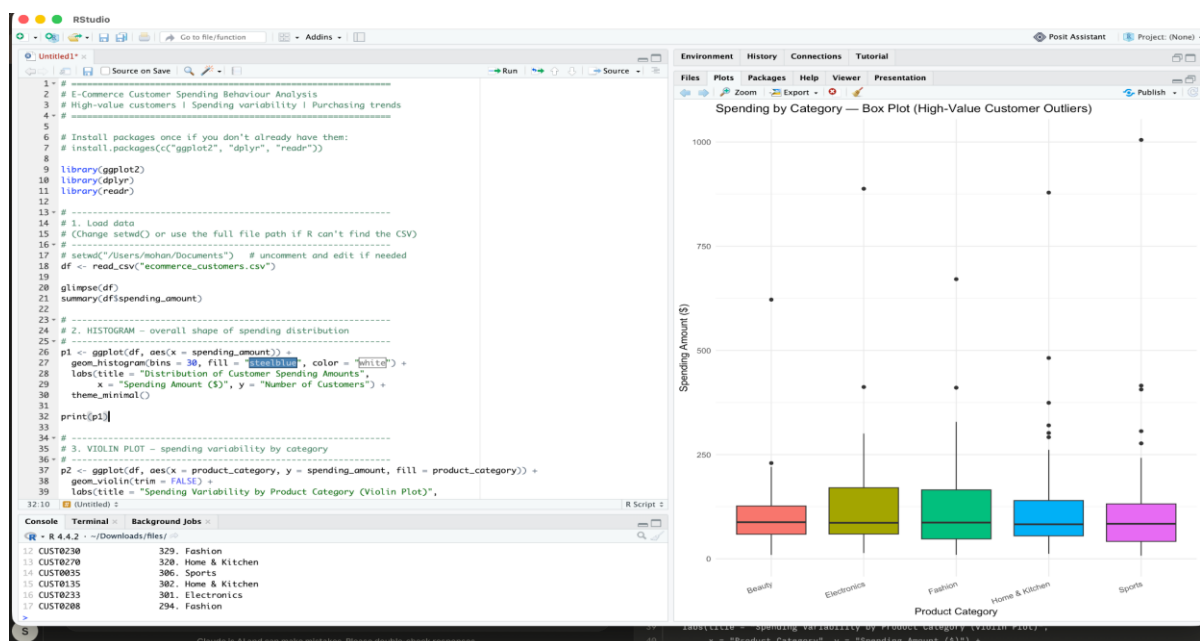
```
ggsave("violin_spending.png", p2, width = 9, height = 5)
```

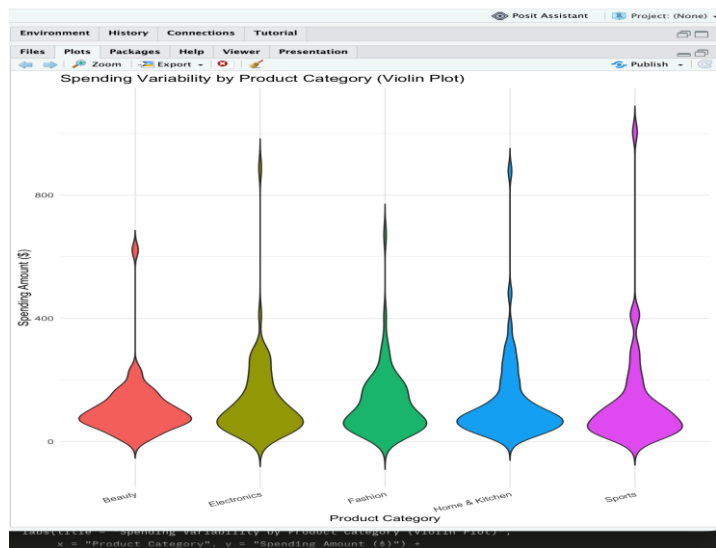
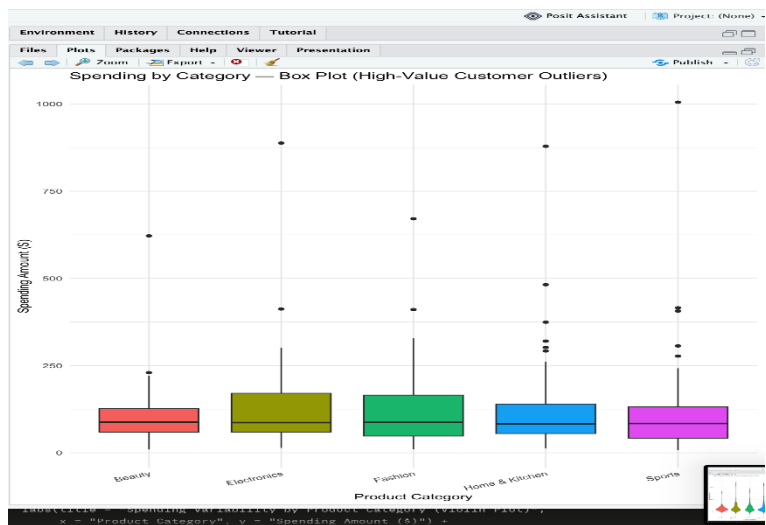
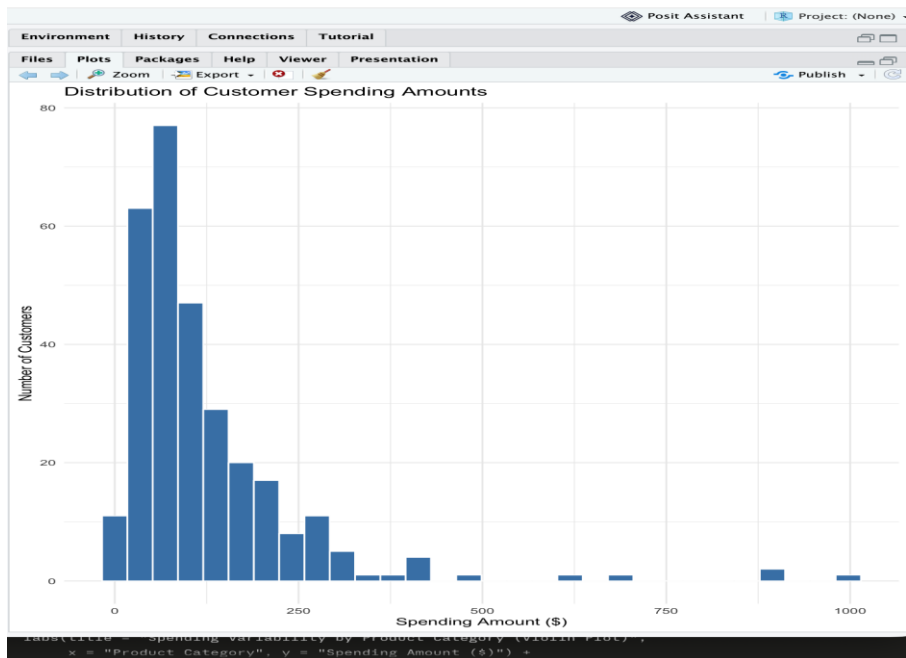
```
ggsave("boxplot_spending.png", p3, width = 9, height = 5)
```

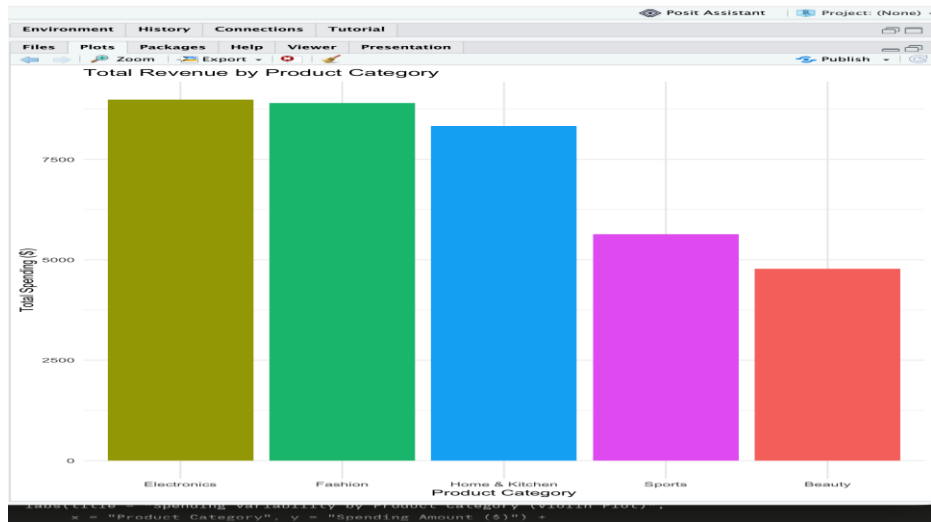
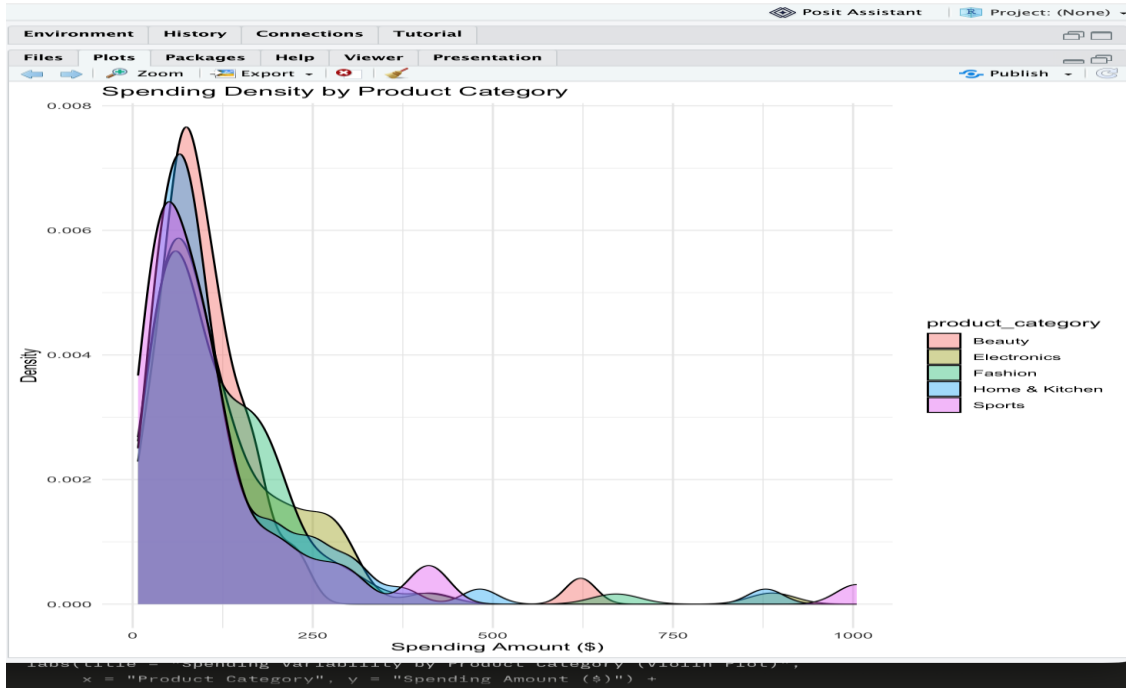
```
ggsave("density_spending.png", p4, width = 8, height = 5)
```

```
ggsave("barchart_revenue.png", p5, width = 8, height = 5)
```

SCREENSHOTS:







3) University Examination Performance Dashboard

A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making. Recommend improvements to the visualization dashboard.

CODE:

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(readr)
```

```
df <- read_csv("exam_performance.csv")
```

```
p1 <- ggplot(df, aes(x = marks)) +  
  geom_histogram(bins = 25, fill = "steelblue", color = "white") +  
  geom_vline(xintercept = 40, linetype = "dashed", color = "red") +  
  labs(title = "Distribution of Exam Marks", x = "Marks", y = "Number of  
Students") +  
  theme_minimal()  
print(p1)
```

```
p2 <- ggplot(df, aes(x = department, y = marks, fill = department)) +  
  geom_boxplot(outlier.color = "red", outlier.size = 2) +  
  labs(title = "Marks by Department", x = "Department", y = "Marks") +  
  theme_minimal() +  
  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust =  
1))  
print(p2)
```

```
p3 <- ggplot(df, aes(x = department, y = marks, fill = department)) +  
  geom_violin(trim = FALSE) +  
  geom_boxplot(width = 0.1, fill = "white", outlier.shape = NA) +  
  labs(title = "Marks Distribution Shape by Department", x = "Department", y =  
"Marks") +  
  theme_minimal() +
```

```

  theme(legend.position = "none", axis.text.x = element_text(angle = 20, hjust =
1))
print(p3)

```

```

p4 <- ggplot(df, aes(x = marks, fill = department)) +
  geom_density(alpha = 0.4) +
  geom_vline(xintercept = 40, linetype = "dashed", color = "red") +
  labs(title = "Marks Density by Department", x = "Marks", y = "Density") +
  theme_minimal()
print(p4)

```

```

class_avg <- df %>%
  group_by(department, class_section) %>%
  summarise(avg_marks = mean(marks), .groups = "drop") %>%
  arrange(desc(avg_marks))

```

```

p5 <- ggplot(class_avg, aes(x = reorder(paste(department, class_section), -
avg_marks),
                        y = avg_marks, fill = department)) +
  geom_col() +
  labs(title = "Top-Performing Class Sections", x = "Department - Section", y =
"Average Marks") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p5)

```

```

outliers <- df %>%
  group_by(department) %>%
  mutate(
    Q1 = quantile(marks, 0.25),
    Q3 = quantile(marks, 0.75),
    IQR_val = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR_val,
    upper_bound = Q3 + 1.5 * IQR_val
  ) %>%
  filter(marks < lower_bound | marks > upper_bound) %>%
  select(student_id, department, class_section, marks) %>%

```

```

arrange(department, desc(marks))
print(outliers)

```

```

pass_summary <- df %>%
  group_by(department) %>%
  summarise(
    total_students = n(),
    pass_count = sum(marks >= 40),
    pass_rate_pct = round(100 * pass_count / total_students, 1)
  ) %>%
  arrange(desc(pass_rate_pct))
print(pass_summary)

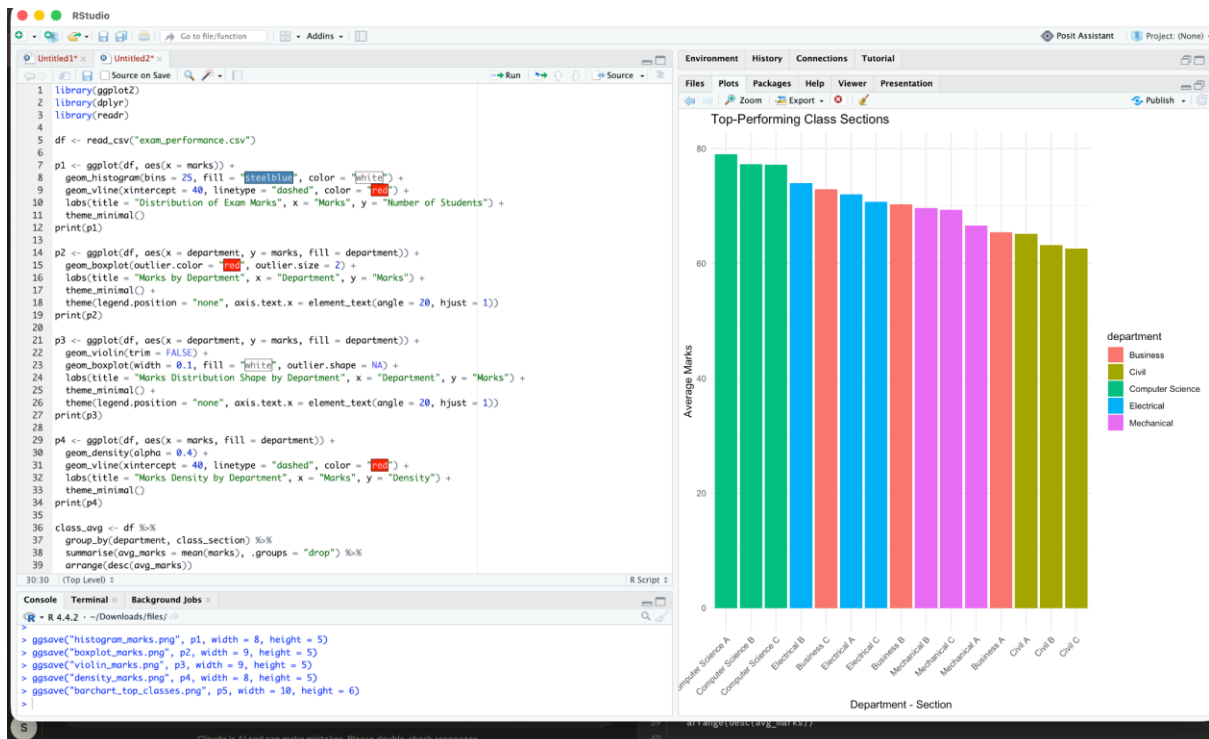
```

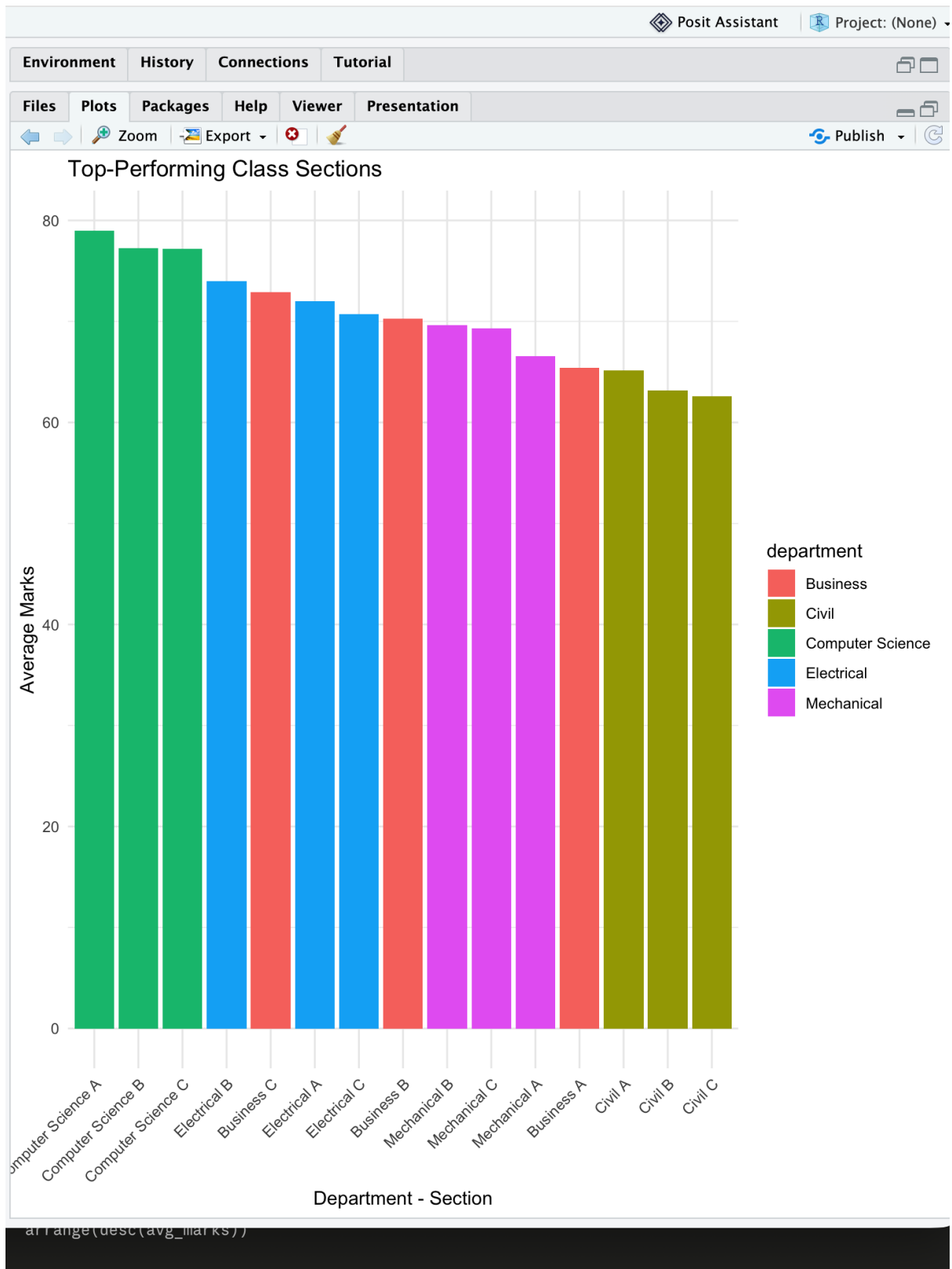
```

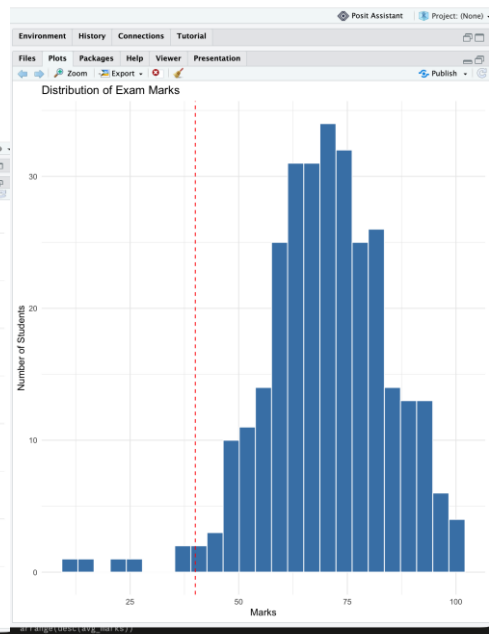
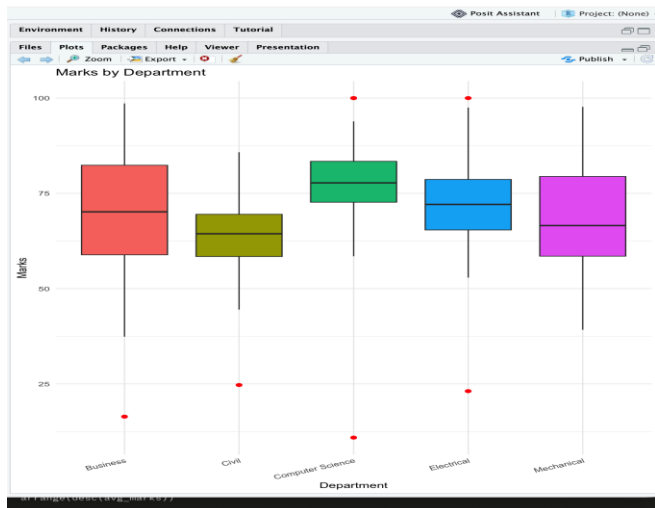
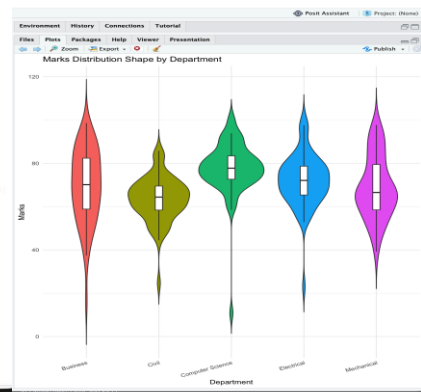
ggsave("histogram_marks.png", p1, width = 8, height = 5)
ggsave("boxplot_marks.png", p2, width = 9, height = 5)
ggsave("violin_marks.png", p3, width = 9, height = 5)
ggsave("density_marks.png", p4, width = 8, height = 5)
ggsave("barchart_top_classes.png", p5, width = 10, height = 6)

```

SCREENSHOTS:







4)Climate and Rainfall Distribution Study

Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental

CODE:

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(readr)
```

```
df <- read_csv("climate_data.csv")
```

```
df$month <- factor(df$month, levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
                                         "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))
```

```
p1 <- ggplot(df, aes(x = month, y = rainfall_mm, fill = city)) +  
  geom_col(position = "dodge") +  
  labs(title = "Monthly Rainfall by City", x = "Month", y = "Rainfall (mm)") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  
print(p1)
```

```
p2 <- ggplot(df, aes(x = rainfall_mm, fill = city)) +  
  geom_density(alpha = 0.4) +  
  labs(title = "Rainfall Distribution by City", x = "Rainfall (mm)", y = "Density")  
+  
  theme_minimal()  
print(p2)
```

```
p3 <- ggplot(df, aes(x = city, y = rainfall_mm, fill = city)) +  
  geom_boxplot(outlier.color = "red", outlier.size = 2) +  
  labs(title = "Rainfall Spread by City (Outliers Highlighted)", x = "City", y =  
"Rainfall (mm)") +
```

```

theme_minimal() +
theme(legend.position = "none")
print(p3)

```

```

p4 <- ggplot(df, aes(x = rainfall_mm)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(title = "Overall Rainfall Distribution", x = "Rainfall (mm)", y =
"Frequency") +
  theme_minimal()
print(p4)

```

```

p5 <- ggplot(df, aes(x = month, y = temperature_c, color = city, group = city)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Monthly Temperature Trend by City", x = "Month", y =
"Temperature (°C)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p5)

```

```

p6 <- ggplot(df, aes(x = month, y = humidity_pct, fill = city)) +
  geom_boxplot() +
  labs(title = "Humidity Variation by Month", x = "Month", y = "Humidity (%)")
+
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
print(p6)

```

```

city_summary <- df %>%
  group_by(city) %>%
  summarise(
    total_rainfall = sum(rainfall_mm),
    avg_temp = round(mean(temperature_c), 1),
    avg_humidity = round(mean(humidity_pct), 1),
    peak_month = month[which.max(rainfall_mm)]
  ) %>%
  arrange(desc(total_rainfall))

```



```
print(city_summary)
```

```
ggsave("bar_monthly_rainfall.png", p1, width = 10, height = 6)
```

```
ggsave("density_rainfall.png", p2, width = 8, height = 5)
```

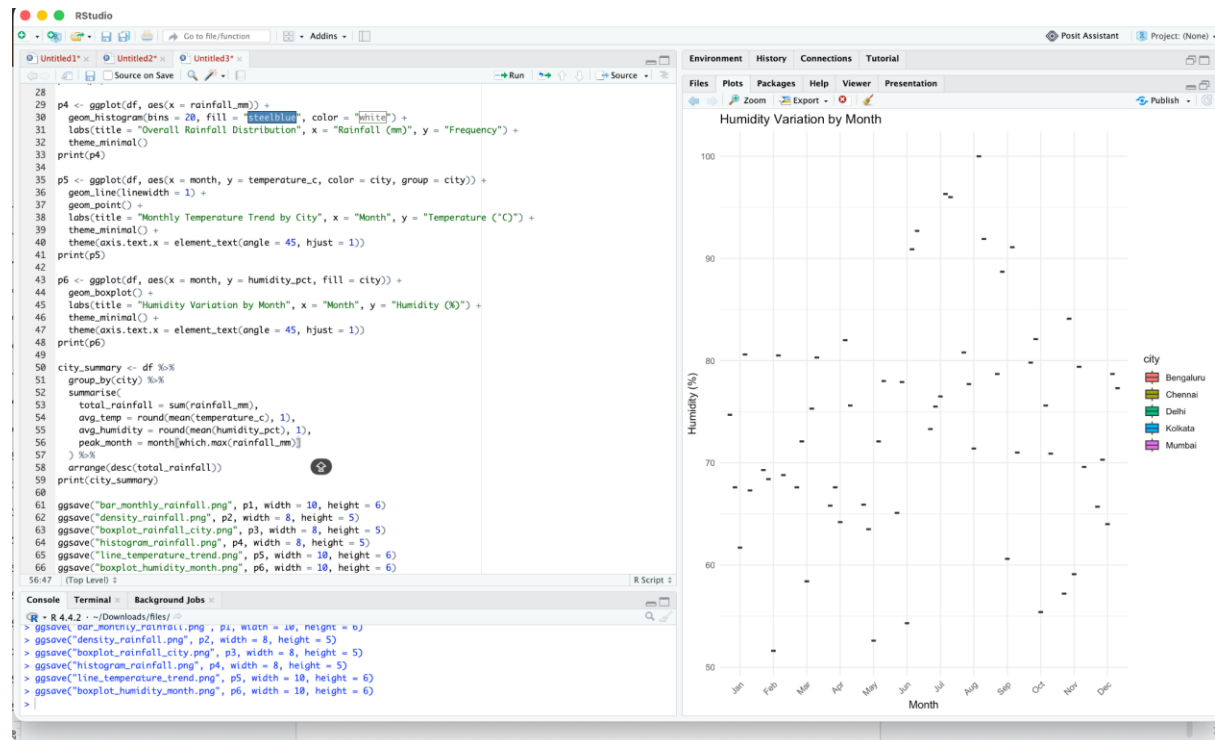
```
ggsave("boxplot_rainfall_city.png", p3, width = 8, height = 5)
```

```
ggsave("histogram_rainfall.png", p4, width = 8, height = 5)
```

```
ggsave("line_temperature_trend.png", p5, width = 10, height = 6)
```

```
ggsave("boxplot_humidity_month.png", p6, width = 10, height = 6)
```

SCREENSHOTS:





5) Bank Loan Risk Assessment:

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard.

CODE:

```
library(ggplot2)
library(dplyr)
library(readr)
```

```
df <- read_csv("loan_risk.csv")
df$repayment_history <- factor(df$repayment_history, levels = c("On-time",
"Late", "Default"))
```

```
p1 <- ggplot(df, aes(x = loan_amount)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Loan Amounts", x = "Loan Amount ($)", y =
"Number of Customers") +
  theme_minimal()
print(p1)
```

```
p2 <- ggplot(df, aes(x = loan_amount, fill = repayment_history)) +
  geom_density(alpha = 0.4) +
  labs(title = "Loan Amount Density by Repayment History", x = "Loan Amount
($)", y = "Density") +
  theme_minimal()
print(p2)
```

```
p3 <- ggplot(df, aes(x = repayment_history, y = loan_amount, fill =
repayment_history)) +
  geom_boxplot(outlier.color = "red", outlier.size = 2) +
  labs(title = "Loan Amount by Repayment History (Outliers Highlighted)",
    x = "Repayment History", y = "Loan Amount ($)") +
  theme_minimal() +
  theme(legend.position = "none")
print(p3)
```

```
df$credit_band <- cut(df$credit_score,
  breaks = c(300, 579, 669, 739, 850),
  labels = c("Poor", "Fair", "Good", "Excellent"))
```

```
default_rate <- df %>%
  group_by(credit_band) %>%
  summarise(
    total_customers = n(),
    defaults = sum(repayment_history == "Default"),
    default_rate_pct = round(100 * defaults / total_customers, 1)
  )
```

```

p4 <- ggplot(default_rate, aes(x = credit_band, y = default_rate_pct, fill =
credit_band)) +
  geom_col() +
  labs(title = "Default Rate by Credit Band", x = "Credit Band", y = "Default Rate
(%)") +
  theme_minimal() +
  theme(legend.position = "none")
print(p4)

```

```

p5 <- ggplot(df, aes(x = credit_score, y = loan_amount, color =
repayment_history)) +
  geom_point(alpha = 0.6) +
  labs(title = "Credit Score vs Loan Amount by Repayment Status",
    x = "Credit Score", y = "Loan Amount ($)") +
  theme_minimal()
print(p5)

```

```

outliers <- df %>%
  group_by(repayment_history) %>%
  mutate(
    Q1 = quantile(loan_amount, 0.25),
    Q3 = quantile(loan_amount, 0.75),
    IQR_val = Q3 - Q1,
    upper_bound = Q3 + 1.5 * IQR_val
  ) %>%
  filter(loan_amount > upper_bound) %>%
  select(customer_id, credit_score, loan_amount, repayment_history) %>%
  arrange(desc(loan_amount))
print(outliers)

```

```

print(default_rate)
ggsave("histogram_loan_amount.png", p1, width = 8, height = 5)
ggsave("density_loan_repayment.png", p2, width = 8, height = 5)
ggsave("boxplot_loan_repayment.png", p3, width = 8, height = 5)
ggsave("barchart_default_rate.png", p4, width = 8, height = 5)
ggsave("scatter_credit_vs_loan.png", p5, width = 9, height = 6)

```

SCREENSHOTS:

